Predicting Recruit 2-Year Starter Probability

Evan Ellington
ICME
Stanford University
eellingt@stanford.edu

Kaitlyn Kwan
Department of Computer Science
Stanford University
kaitkwan@stanford.edu

Megan Liu
Department of Economics
Stanford University
meganl@stanford.edu

Abstract

This project predicts whether a college football recruit will become a starter within two years. Using recruiting and player data from CollegeFootballData.com, we built a filtered dataset of 7,136 players and used features such as star rating, recruiting rating, rating difference compared to average at that position, height, weight, position, and conference to model early starting outcomes. We compared a simple baseline rule—predicting starter status for 4- and 5-star recruits—to Logistic Regression, Random Forest, XGBoost, and a stacked ensemble, all evaluated on a 70/30 train-test split. We find that the machine learning models outperform the baseline, with Random Forest and XGBoost performing best overall.

1 Introduction

1.1 The Problem and Our Ideas

The advent of Name, Image, and Likeness (NIL) in college athletics has transformed recruiting from a traditional talent search into a high-stakes portfolio allocation problem. Because programs operate under strict constraints—including limited roster spots, finite financial reserves, and developmental bandwidth—every recruit represents a significant investment made under conditions of meaningful uncertainty. Consequently, team managers must function as portfolio managers, judging which prospects are most likely to justify their acquisition cost by contributing early and improving overall team outcomes.

In this environment, the speed of a player's transition from prospect to contributor is a critical metric of investment success. While long-term development remains a secondary goal, the immediate pressures of the modern game prioritize recruits capable of making an instant impact. This project addresses this priority by focusing on a specific measure of early contribution: whether a recruit earns a starting role within his first two years. By utilizing player usage—defined as the share of offensive plays in which a player is directly involved—we analyze how effectively standard recruiting data can forecast which athletes translate their raw physical profile into on-field opportunities.

To operationalize this, we define the binary indicator `starter_by_year2` based on whether a player's maximum usage during his initial two seasons exceeds a calculated threshold. This threshold-based approach allows for a robust classification of whether a recruit earned a meaningful role within the program's immediate competitive window.

1.2 Past Literature

Prior research shows that recruiting rankings in college football convey meaningful information, though most studies focus on team-level outcomes rather than individual early contribution. Bergman and Logan (2016) find that higher-quality recruiting classes are associated with stronger team performance even after controlling for school fixed effects, suggesting recruit ratings capture more than the advantages of already-strong programs [Bergman and Logan, 2016]. They later show that

38th Conference on Neural Information Processing Systems (NeurIPS 2024).

recruit quality is also linked to football-specific revenue, indicating that recruiting evaluations are both competitively and economically informative [Bergman and Logan, 2020]. While this literature establishes the value of recruiting rankings, it leaves open how quickly highly rated recruits contribute after arriving on campus.

Early contribution is distinct from long-run success. Players may eventually become valuable while failing to earn major playing time early due to injuries, depth-chart competition, coaching decisions, scheme fit, or physical development. Evidence across sports suggests early talent signals are informative but imperfect predictors of future outcomes. Barth et al. (2022) show that predictors of junior success differ from those of elite senior performance, and Barth et al. (2024) find that early talent promotion does not consistently translate into later achievement [Barth et al., 2022, 2024]. These findings highlight the importance of studying early contribution separately from long-run outcomes.

Building on this literature, our project examines whether a recruit becomes an early contributor within two years. We measure early contribution using usage and starter status by year two, shifting the focus from overall team success or eventual outcomes to the timing of on-field opportunity. This perspective is relevant for roster construction, player development, and evaluating the immediate impact of recruiting classes.

Our approach also connects to the growing use of machine learning in sports analytics. Machine learning can capture nonlinear relationships and interactions that simpler models may miss [Pietraszewski et al., 2025], which is important because the relationship between recruit characteristics and early contribution may vary by position, conference, and physical profile. By comparing a recruiting-based baseline with logistic regression and tree-based machine learning models, our project applies predictive methods to a meaningful college football outcome.

2 Methods

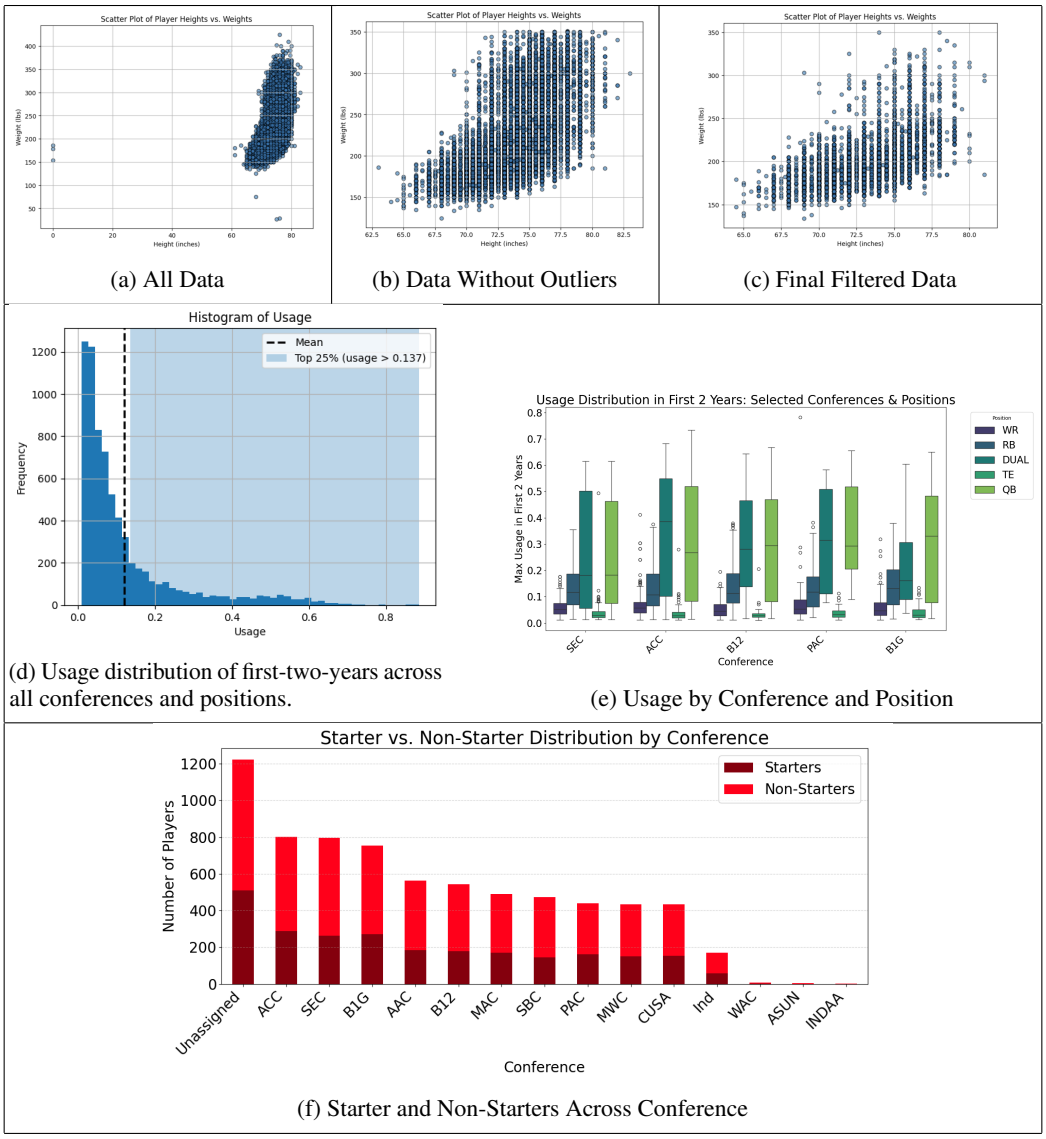
2.1 The Dataset

Our dataset was built using recruiting and player data from CollegeFootballData.com spanning between 2015 and 2025. The recruiting data includes player-level information such as recruiting year, position, recruiting rating, height, and weight, while the player data provides college usage statistics that allow us to measure on-field contribution. The usage of a player is defined as the percentage of plays in which a player was directly involved, including passing downs, standard downs, passes, rushes, as well as first, second and third downs. These are all compiled into an overall usage score, which is what we use as our primary indicator player usage. By combining the recruiting and player data, we are able to connect recruiting characteristics to whether a player contributes in his college career.

To prepare the data for analysis, we first merged the recruiting and player datasets by player name after standardizing names across both sources. Players who appeared in the college player data but could not be matched to a recruiting profile were excluded(Figure 1a). We addressed outliers in physical characteristics by filtering extreme height and weight values. There were 49 players that were found to be outliers, and none of them were determined to be a starter via the process outlined below, so we felt confident in removing them from the dataset(Figure 1b). There were also a large amount of players that were never successfully recruited to a team, they were denoted as `Did Not Play` in the `team` column of the data. After confirming all these players did not have any recorded usage and were therefore not a starter, they were filtered out of the dataset(Figure 1c). We calculated how many years had passed since each player's recruiting year and focused on outcomes within the first two years. To engineer the starter measure, we first calculated each player's maximum usage during his first two years, denoted `max_usage_first2yrs` (Figure 1d). We then defined `starter_by_year2` as an indicator equal to 1 if `max_usage_first2yrs` was over 0.137 (corresponding to the top 25% of recruits' usage in figure 1d), and 0 otherwise. Additionally, we created a feature called `rating_diff`, defined as the difference between a player's recruiting rating and the average rating of all other players at the same position. This feature captures a recruit's relative standing within their positional group, allowing the model to measure how much stronger or weaker a player is compared to typical recruits at that position rather than relying only on absolute ratings.

Figure 1e shows the distribution of usage in the first two years across selected conferences and positions. The figure highlights substantial differences across player roles, with quarterbacks and dual-threat players generally displaying higher usage than positions such as wide receivers or tight ends. Figure 1f shows how the resulting starter and non-starter classifications are distributed across conferences in the cleaned sample. The `Unassigned` conference consists of a variety of smaller conferences, such as the Big Sky conference, Missouri Valley Football Conference, Ivy League, Ohio Valley Conference, etc.

Figure 1: Data Visualization Figures



(a) All Data

(b) Data Without Outliers

(c) Final Filtered Data

(d) Usage distribution of first-two-years across all conferences and positions.

(e) Usage by Conference and Position

(f) Starter and Non-Starters Across Conference

Figure 1

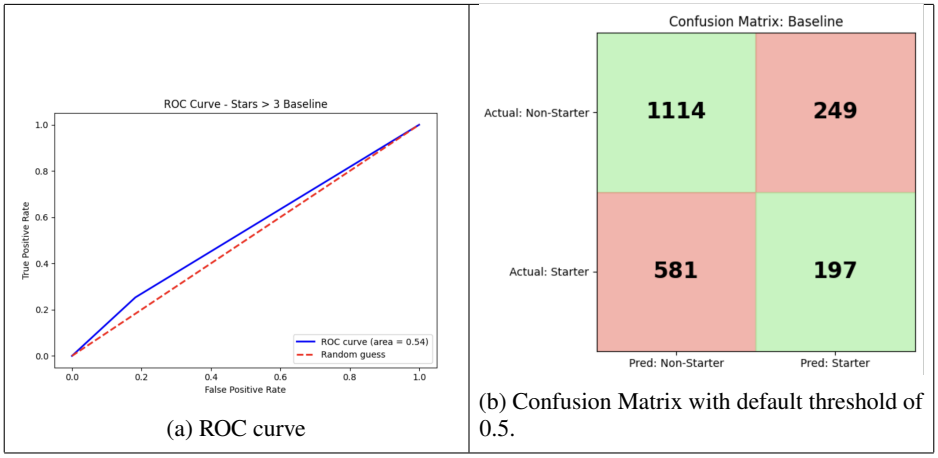
2.2 Our Technical Methods

2.2.1 Baseline Model

We began with a simple baseline classifier based only on recruiting stars. This baseline provides an interpretable benchmark before introducing more flexible machine learning models. Specifically, we predicted that a recruit would become a starter by year two if he was a 4- or 5-star recruit, In other words, classifying players with `stars > 3` as starters.

We split the data into training and test sets using a 70/30 split and measured baseline performance using accuracy, ROC AUC (figure 2a), a classification report, and a confusion matrix(figure 2b). This benchmark is useful because star ratings are one of the most common summary measures of recruit quality. As a result, it provides a natural comparison point for judging whether more advanced models add predictive value beyond a simple recruiting heuristic.

Figure 2: Baseline Model Figures



(a) ROC curve

(b) Confusion Matrix with default threshold of 0.5.

Figure 2: Baseline Model evaluation figures.

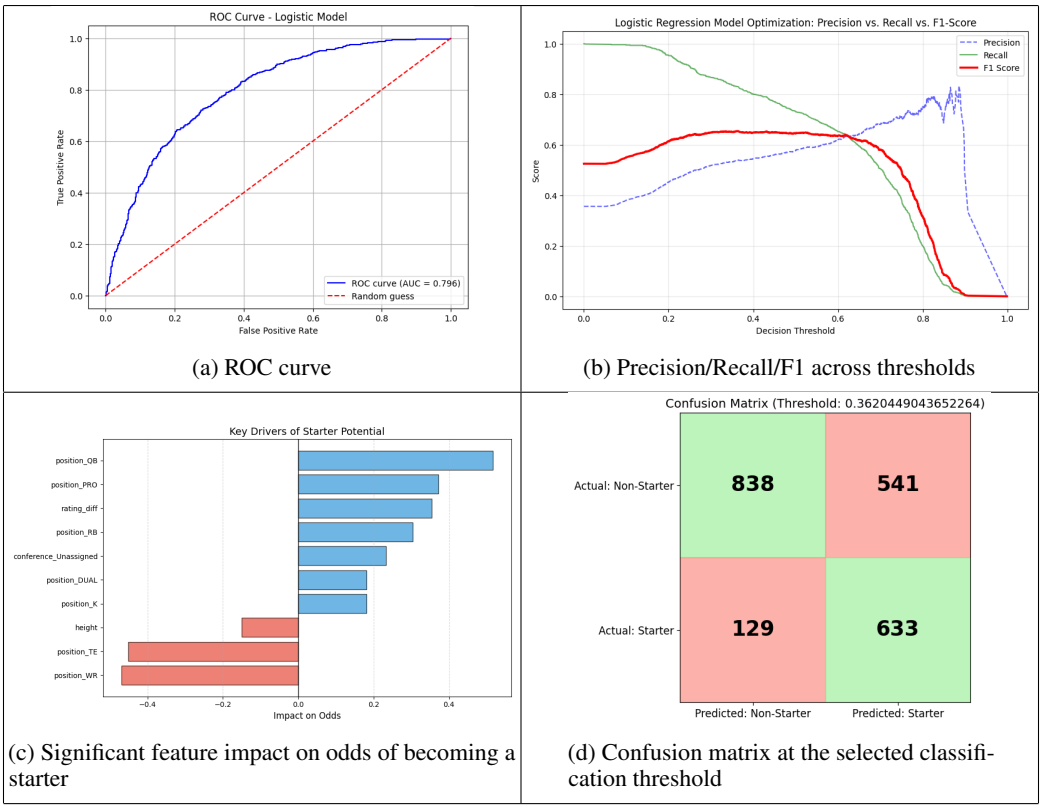
2.2.2 Logistic Regression Model

We next used a logistic regression model and unlike the baseline model, which relies only on recruiting stars, the logistic model allows the probability of becoming a starter to vary with multiple recruit characteristics simultaneously.

Before fitting the model, we examined multicollinearity among the quantitative features. We dropped the `stars` feature since it was highly correlated with `rating` and `rating` provides a better continuous range than `stars` for modeling. Then we used the variance inflation factor (VIF) calculations to see that `rating` and `rating_diff` showed strong collinearity, so we did A/B testing on the predictive performance if either feature and retained `rating_diff` as the higher performer and excluded `rating`. Our final logistic regression model used `position`, `height`, `weight`, `conference`, and `rating_diff` as predictors. Categorical variables such as `position` and `conference` were converted into dummy variables, and all features were standardized before estimation.

We split the data into training and test sets using a 70/30 split with stratification on the outcome variable to preserve the starter share across samples. The model was estimated using the `liblinear` solver with `class_weight='balanced'` in order to account for class imbalance between starters and non-starters. We then evaluated performance on the test set using accuracy, ROC AUC (figure 3a). We examined how precision, recall, and F1 varied across classification thresholds (figure 3b). To complement these performance metrics, we also examined the fitted logistic coefficients to identify which features were most strongly associated with starter probability. Figure 3c highlights the significant coefficients in the model and how they impact the odds of a recruit being a starter by year 2.

Figure 3: Logistic Regression Figures



(a) ROC curve

(b) Precision/Recall/F1 across thresholds

(c) Significant feature impact on odds of becoming a starter

(d) Confusion matrix at the selected classification threshold

Figure 3

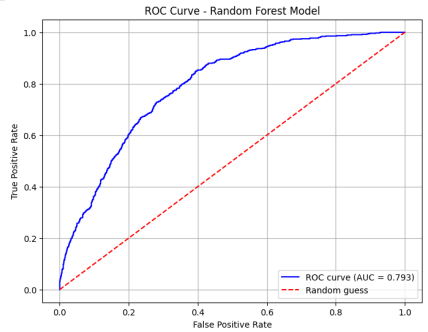
2.2.3 Random Forest Model

We next used a random forest model to allow for more flexible nonlinear relationships between recruit characteristics and early starter outcomes. Unlike logistic regression, which imposes a linear structure on the log-odds of becoming a starter, random forest can capture interactions and threshold effects automatically by combining predictions from many decision trees.

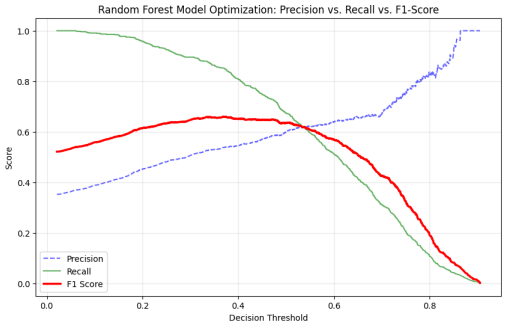
To construct the model, we included recruit characteristics: `rating`, `position`, `height`, `weight`, `conference`, and `rating_diff`. Similar to logistic regression, `position` and `conference` were converted into dummy variables using one-hot encoding. We again split the data into training and test sets using a 70/30 split.

We performed a random grid search over a set of hyperparameters, varying tree depth and minimum split size, and found that using 500 trees, a maximum depth of 15, and a minimum split size of 5 gave us optimal performance on the data. We also set `class_weight="balanced"` to account for the imbalance between starters and non-starters in the dataset. We evaluated the model's performance using test-set accuracy and ROC AUC (figure 4a). Again, we plotted precision, recall, and F1 score as functions of the decision threshold and identified the threshold that maximized F1 (figure 4b). To better understand which variables matter most in the model, we computed permutation importance on the test set (figure 4c). Rather than relying only on impurity-based importance measures, permutation importance evaluates how much predictive performance falls when a feature is randomly shuffled, making it a more interpretable measure of contribution. We focus especially on the quantitative features to assess the relative importance of athletic and recruiting characteristics in predicting early starter potential. Finally, we computed the SHAP values of the quantitative features (figure 4e) to show both the relative importance of each feature and the direction of its effect on the model's predictions. The random forest model improves substantially on the simple star-based baseline and captures more flexible nonlinear relationships in early starter prediction.

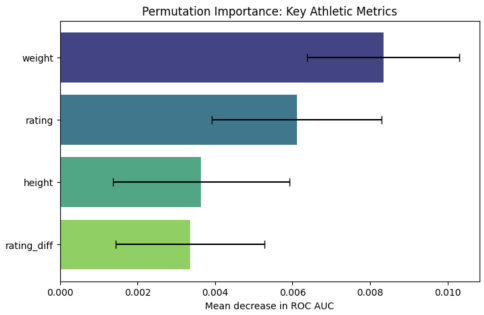
Figure 4: Random Forest Figures



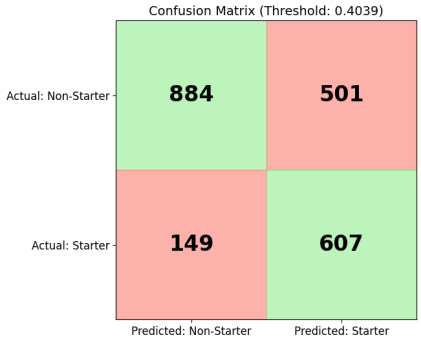
(a) ROC curve



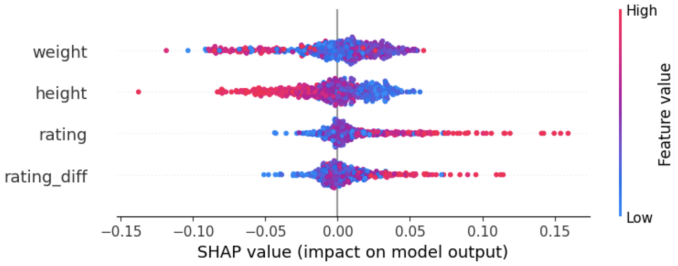
(b) Precision/Recall/F1 across thresholds



(c) Permutation importance of quantitative features



(d) Confusion matrix at the selected classification threshold



(e) SHAP summary plot showing the impact of key features on model predictions

Figure 4

2.2.4 XGBoost Model

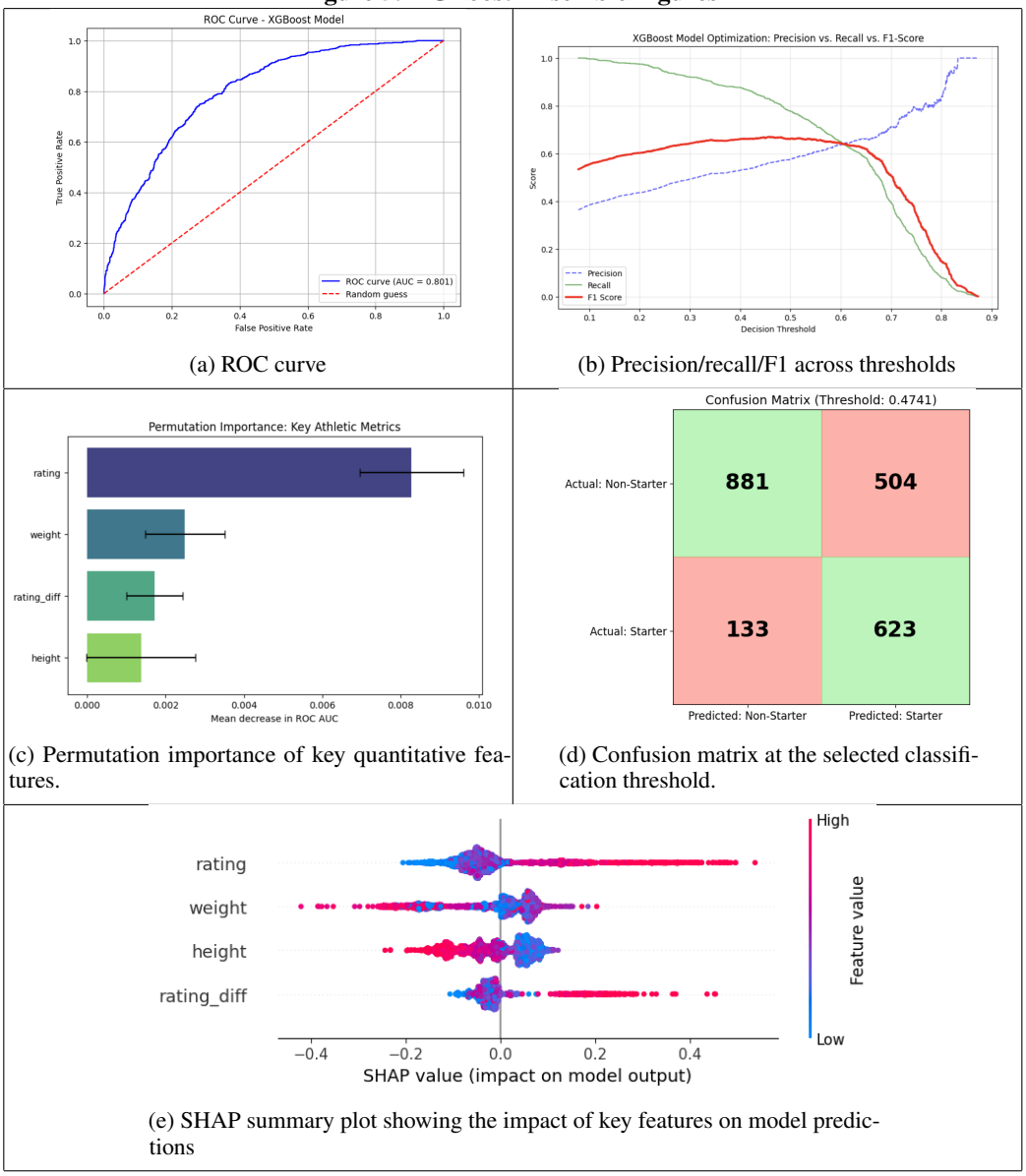
We next made an XGBoost model as a more flexible boosting-based alternative to both logistic regression and random forest. XGBoost is well suited to this setting because it can capture nonlinearities and interactions while sequentially improving prediction errors across many trees.

To construct the model, we again, one-hot encoded the categorical position and conference variables and used a 70/30 split. To account for the imbalance between starters and non-starters, we set `scale_pos_weight` equal to the ratio of negative to positive examples in the training data. We ran a randomized hyperparameter search over tree depth, learning rate, number of estimators, and subsampling parameters and ended up using 100 trees, a learning rate of 0.05, a maximum depth of 3, and subsampling rates of 0.7. We also enabled early stopping based on performance on the test set in order to reduce overfitting.

We evaluated the XGBoost model using test-set accuracy and ROC AUC (figure 5a). We again plotted precision, recall, and F1 score across thresholds and identified the threshold that maximized F1 (figure 5b) with the corresponding confusion matrix (figure 5d) as well as the permutation importance on the test set (figure 5c) and the SHAP values (figure 5e).

The XGBoost model provides another step toward a more flexible predictive framework. Relative to the simpler baseline and logistic models, it is able to capture more complex patterns in the data, while the threshold and confusion-matrix analysis illustrates how model evaluation still depends on the tradeoff between identifying more true starters and limiting false positives.

Figure 5: XGBoost Ensemble Figures



(a) ROC curve

(b) Precision/recall/F1 across thresholds

(c) Permutation importance of key quantitative features.

(d) Confusion matrix at the selected classification threshold.

(e) SHAP summary plot showing the impact of key features on model predictions

Figure 5

2.2.5 Stacked Ensemble

Finally, we implemented a stacked ensemble classifier to investigate whether combining the predictions of our three base models (logistic regression, random forest, and XGBoost) could yield improvements beyond what any individual model achieved. The motivation for stacking is that if base

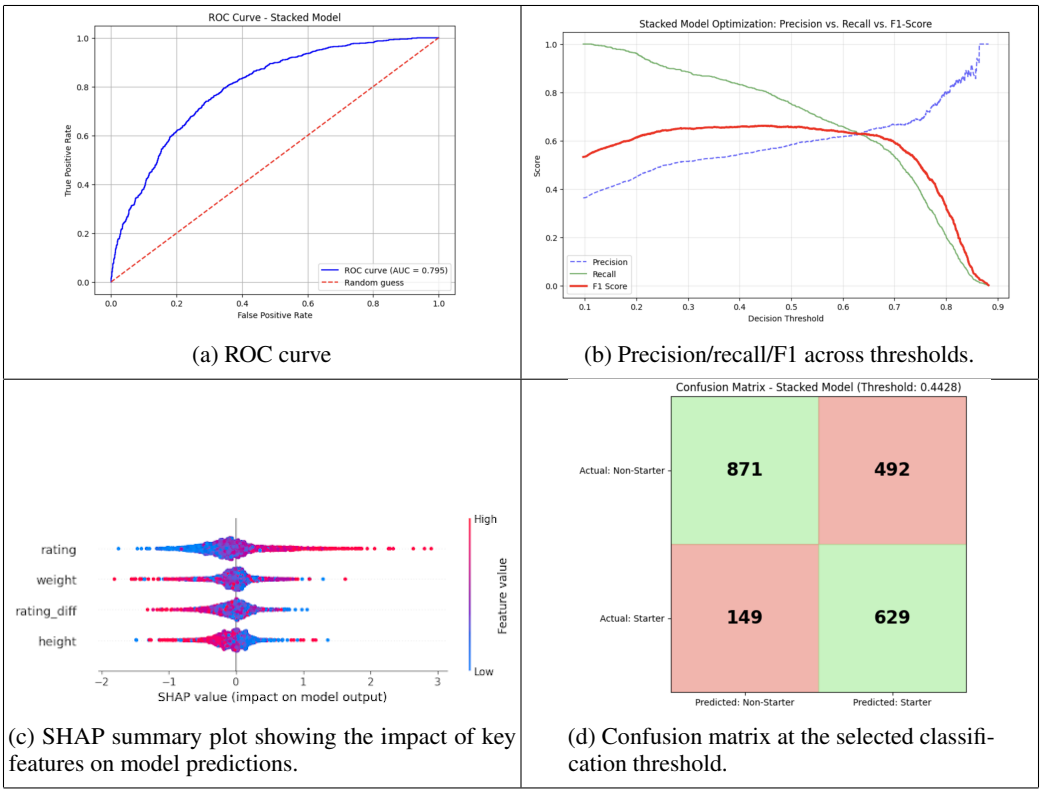
models make systematically different errors, a meta-learner trained on their combined outputs can exploit complementary signal to improve overall predictive performance. This is especially appealing in a setting where individual models have plateaued at similar performance levels.

To construct the stacked model, we generated out-of-fold predictions from each base model using 5-fold cross-validation on the training set. Training the meta-learner on held-out predictions in this way prevents data leakage, ensuring the stacking layer never sees in-sample fitted values. The out-of-fold predicted probabilities from all three base models were then concatenated into a new feature matrix, which served as input to the meta-learner. We used logistic regression as the meta-learner, as it provides a transparent and regularized method for combining base model outputs.

The base models retained the same hyperparameters as their standalone versions, and the full pipeline was applied to the test set by first generating base model predictions before passing them to the meta-learner for final probability estimate. As with the individual models, we used a 70/30 train-test split with a fixed random seed to ensure comparability between models when run individually and stacked.

We evaluated the stacked model using the same metrics as our base models: test-set accuracy, ROC AUC (figure 6a), precision, recall, and F1 score at an optimized classification threshold (figure 6b). To further understand how the meta-learner weighted its inputs, we computed SHAP values on the stacked model's XGBoost component (figure 6c), allowing us to assess the relative contribution for each base model's predictions to the final output.

Figure 6: Stacked Ensemble Figures



(a) ROC curve

(b) Precision/recall/F1 across thresholds.

(c) SHAP summary plot showing the impact of key features on model predictions.

(d) Confusion matrix at the selected classification threshold.

Figure 6

3 Results

	Baseline	Logistic Regression	Random Forest	XGBoost	Stacked
Starter Predictions					
Precision	0.44	0.37	0.56	0.57	0.56
Recall	0.25	0.47	0.82	0.83	0.83
F1	0.32	0.41	0.67	0.67	0.67
Non-Starter Predictions					
Precision	0.66	0.64	0.86	0.86	0.87
Recall	0.82	0.54	0.63	0.64	0.64
F1	0.73	0.58	0.73	0.73	0.74

Table 1: Model performance for predicting starter and non-starter outcomes across different models.

Overall, logistic regression improved on the baseline but remained weaker than the tree-based models. The largest gap appeared in predicting starters: logistic regression produced a much lower Starter F1 than both random forest and XGBoost, suggesting that a linear model struggled to capture the relationships in the data. In contrast, random forest and XGBoost performed best overall and were nearly tied on Starter F1 at around 0.67, indicating a better balance of precision and recall for the more difficult class.

Between the two, XGBoost had a slight edge in starter precision and recall—particularly recall—meaning it identified true starters slightly more effectively, while random forest performed similarly overall. On the non-starter side, both tree-based models again performed almost identically and outperformed logistic regression. These results suggest that tree-based models better capture nonlinear relationships among features such as rating, rating difference, height, weight, position, and conference. However, even the best models were more accurate at identifying non-starters than starters, highlighting the greater difficulty of predicting starters.

The stacked ensemble was intended to exceed the Starter F1 plateau of 0.67 observed in both random forest and XGBoost. Instead, it matched the F1 of its base models and tied XGBoost on recall. This outcome appears largely due to the very high prediction similarity between random forest and XGBoost (0.9613), leaving little complementary error for the ensemble to exploit. For non-starters, the stacked model achieved the highest precision and F1, though the gains were marginal. Overall, these results suggest that further tuning of individual models may be more productive than ensembling when optimizing for starter prediction F1.

4 Conclusion and Future Work

This project has several limitations. Our models rely primarily on public recruiting data—ratings, height, and weight—which are only proxies for actual high school performance and do not capture finer measures of skill, production, or scheme fit. The modeling sample is restricted by data availability, as many recruits lacked meaningful college participation within the two-year window. Additionally, our analysis does not account for redshirting, so some players classified as non-starters may have intentionally delayed playing to develop. Finally, the `starter_by_year2` variable approximates starter status using maximum usage rather than official starts, which may not fully reflect roster dynamics, injuries, transfers, or coaching decisions.

Future work could address these limitations and extend the framework in several directions. Incorporating NIL data would allow modeling player value more directly, linking expected compensation to early on-field contribution. Extending the framework to transfers could help predict immediate impact in new programs, reflecting the growing importance of the transfer portal. Position-specific models could improve predictive precision, as the factors influencing early contribution differ across positions. More broadly, richer data on player development and team context—such as injuries, coaching changes, depth-chart competition, offensive schemes, and high school production—would enhance both the realism and practical utility of the models in evaluating early player impact.

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A Appendix / supplemental material

Optionally include supplemental material (complete proofs, additional experiments and plots) in appendix. All such materials **SHOULD be included in the main submission.**